

Video Transcripts

Week 2.1: Fundamentals of Generative AI – How It Creates Content

Video 1: What “Generation” Really Means

In business settings, artificial intelligence behaves very differently depending on the task it is designed for. Some systems analyse data to estimate risk, likelihood, or category, while others actively produce the materials that professionals use in their daily work. This distinction becomes clear when we observe AI within real workflows. Let us begin with predictive AI.

Predictive AI supports decision-making by analysing historical patterns and estimating outcomes. Insurance firms use it to assess claim risk and recommend investigation thresholds, airlines apply it to optimise dynamic pricing based on booking behaviour and seasonality, and manufacturing companies rely on predictive maintenance models to estimate the likelihood of equipment failure and schedule servicing. In each case, the output is a likelihood score, risk band, or forecast. The system informs human decisions, but it does not produce new documents, messages, or artefacts.

Generative AI, by contrast, produces content that professionals would otherwise create manually. It generates structured text, documents, explanations, and code in response to instructions. Instead of estimating what might happen, it creates outputs that can be directly reviewed, edited, and used. For example, a compliance team responding to a lengthy regulatory circular can prompt the system to create a one-page internal summary highlighting key changes and required actions. Similarly, a finance team during month-end close can generate concise variance explanations for leadership, while a procurement manager can draft a formal follow-up email to a supplier regarding delayed deliveries. In each case, the AI produces complete, well-structured content ready for refinement.

A large-scale example of this can be seen in the legal industry. In 2023, the global law firm Allen & Overy deployed a generative AI system across the organisation to support drafting and legal research. During the rollout, around 3,500 lawyers used the system, generating approximately 40,000 queries as part of their everyday work. Usage data showed that about one in four lawyers used the tool daily, and nearly 80% used it at least once a month. One of the primary use cases was drafting first-pass client memos and legal documents. Lawyers could prompt the system to generate an initial advisory note summarising regulatory implications, which they would then review, refine, and approve before sharing with clients.

Across these examples, the distinction is consistent. Predictive AI produces assessments that guide decisions, whereas generative AI produces artefacts that professionals can directly use, such as internal notes, financial explanations, formal communications, and legal drafts. This capability is powered by large language models built on Transformer architectures. During training, these models learn patterns from vast datasets by adjusting neural network weights to minimise prediction error. Text is converted into embeddings and processed through multiple neural network layers using self-attention mechanisms, enabling the model to generate outputs by predicting the most likely sequence of words based on context.

Because these outputs are probabilistic, generative AI can occasionally produce incorrect or incomplete information or reflect biases present in its training data. For this reason, human review remains essential. In practice, generative AI is used to create first drafts and structured outputs, while professionals retain responsibility for accuracy, judgement, and accountability. In summary, through examples such as regulatory notes, financial commentary, supplier communication, and legal drafting

at scale, “generation” becomes clear: it is the creation of usable business content, not the prediction of outcomes.

Video 2: Autoregressive Generation — The Next Token Principle

To truly understand generative AI, particularly large language models, we need to examine a foundational mechanism that underpins everything they produce. This mechanism is known as autoregressive generation. It refers to the way outputs are constructed sequentially, one token at a time, based entirely on what has already been generated. The system does not plan an answer in advance or determine an endpoint; instead, it predicts what comes next, adds it to the sequence, and repeats this process until the output is complete.

A token is not always a full word; it can be part of a word, a number, punctuation, or a symbol. When a user enters a prompt, the model evaluates the probability of every possible next token given the current context, selects one, and immediately proceeds to predict the next. This process is known as next-token prediction. A well-known real-world example of this mechanism is Gmail Smart Compose. When introduced by Google, it suggested short phrases as users typed emails, helping users save over one billion characters per week within months of launch and reducing typing effort by around 15 per cent for those who enabled it.

The principle behind Smart Compose is straightforward. As a user begins typing a phrase such as “Looking forward to...”, the system predicts likely continuations such as “hearing from you” or “your response” based on common patterns in professional email communication. It does not understand intent or context in a human sense; rather, it predicts the most probable sequence of tokens based on patterns learned from large datasets. Large language models operate on the same principle, but at a much greater scale. Instead of completing short phrases, they generate entire emails, summaries, reports, or code by repeatedly predicting the next token, with the appearance of intelligence emerging from the scale of training data and contextual depth.

This sequential process also explains why generative AI is probabilistic rather than deterministic. The model does not retrieve a single correct answer; instead, it samples from a range of likely continuations. As a result, the same prompt can produce slightly different outputs, and even small changes in phrasing can lead to noticeably different results. From a business perspective, this creates both advantages and risks. The advantage lies in rapid variation—marketing teams can generate multiple campaign messages quickly, and product teams can explore alternative descriptions without starting from scratch.

However, the probabilistic nature of the system also introduces uncertainty. Because the model optimises for plausibility rather than verifying facts in real time, it can produce confident but incorrect statements, often referred to as hallucinations. Additionally, autoregressive generation is constrained by a context window, meaning the model can only consider a limited amount of information at once, including the prompt, prior conversation, and any provided material. If critical information falls outside this window, it no longer influences the output, which can result in responses that are coherent but incomplete or misaligned.

Autoregressive generation also explains the speed of generative AI. Rather than reasoning step by step like a human, the model performs rapid probability calculations repeatedly, allowing it to generate long outputs in seconds. For business leaders, the key takeaway is clear: generative AI does not think or plan—it predicts what comes next based on patterns and context. Clear prompts help narrow the probability space and improve reliability, while human review remains essential, as outputs should be treated as drafts rather than definitive answers.

Video 3: Tokenisation — How Text Becomes Tokens

To use generative AI effectively in business, it is essential to understand tokens, as they directly influence cost, response time, and how much information a model can process at once. When AI systems feel expensive or inconsistent, the underlying issue is often poor token management rather than model capability. Tokens are not the same as words. Large language models break text into smaller units, including full words, sub-words, punctuation, numbers, and even spaces. As a result, 1,000 words do not equal 1,000 tokens. In English, one token averages roughly three-quarters of a word, although this varies depending on the structure of the text.

This effect becomes more pronounced with business vocabulary. Terms such as “hyperautomation,” “blockchain-enabled,” or “micro-segmentation” often split into multiple tokens, while acronyms behave inconsistently. Names and culturally specific phrases, such as “Raghunandan,” “Chandrasekhar,” or “Aatmanirbhar Bharat,” may also break into several tokens, increasing usage in ways that are not immediately obvious. This matters because most enterprise AI platforms charge based on tokens, covering both input and output. Longer prompts, repeated context, and unbounded responses increase cost, while higher token usage also requires more computation, leading to increased latency at scale.

Tokens also define the context window, which acts as the model’s working memory across the prompt, conversation history, and any supporting documents. If the total number of tokens exceeds this limit, earlier content is removed. This can appear as inconsistency or “forgetfulness”, but it is simply the result of a fixed memory capacity. At this stage, however, text has only been broken into pieces, and tokens themselves carry no inherent meaning. For the model, each token is simply an identifier within a vocabulary.

Meaning is introduced through embeddings. An embedding converts each token into a numerical vector, typically consisting of hundreds or thousands of values. These vectors position words within a high-dimensional space, where distance represents similarity. For instance, tokens related to finance cluster closer together than those related to healthcare, and words such as “revenue” and “profit” occupy nearby regions. This structure is not manually defined; it is learned during training. In simple terms, tokenisation breaks text into units, while embeddings transform those units into meaningful numerical representations.

Another critical component is positional encoding. Unlike humans, a model does not naturally understand sequence or order. For example, the sentence “Margin improved after costs fell” conveys a different meaning from “Costs fell after margin improved.” To capture this distinction, models apply positional encoding, adding numerical signals to embeddings so the model can recognise the order of tokens within a sequence. By this point, each word in a sentence has been converted into a token, transformed into a vector, and tagged with positional information, enabling the model to begin analysing relationships.

From a business perspective, this stage of text representation plays a key role in both cost efficiency and output reliability. Clear and structured prompts help reduce unnecessary token usage, summarising inputs prevents exceeding the context window, and specifying response length avoids uncontrolled expansion. At enterprise scale, these small optimisations accumulate into significant cost savings and more consistent performance. In summary, tokenisation converts language into units, embeddings provide meaning, and positional encoding preserves order—together transforming text into a mathematical structure that a neural network can process.

Video 4: Transformer Architecture

To understand why generative AI advanced so rapidly after 2017, it helps to begin with a key limitation of earlier systems. While they could process text, they struggled with long-range context, meaning that as sentences became longer, coherence and accuracy often degraded. Most of these earlier models were built using recurrent neural networks, or RNNs, which processed language sequentially, one word at a time. Information was passed along like a chain, step by step. Although LSTMs improved memory to some extent, they still faced challenges with very long dependencies and were slow to train because each step relied on the one before it.

In 2017, researchers introduced a fundamentally different approach in the paper *“Attention Is All You Need.”* Instead of reading text sequentially, the new model examined all tokens at once and calculated relationships between them. This architecture, known as the Transformer, processes sequences in parallel, significantly reducing training time. However, the true breakthrough lies in a mechanism called self-attention, which allows each token to compare itself with every other token in a sentence and assign weights based on their relevance to one another.

For example, consider the sentence: “When the supplier submits the invoice after delivery, they must include proof of compliance.” To interpret the word “they,” the model assigns a stronger connection to “supplier” than to “delivery” or “invoice.” This is achieved through learned numerical relationships rather than explicit grammar rules. Self-attention enables direct links between distant words, improving translation accuracy, summarisation quality, and coherence in longer documents. In a sentence such as “Although revenue increased, margins declined due to logistics costs,” attention mechanisms preserve both ideas rather than oversimplifying the meaning.

Transformers replaced RNNs not only because they improved accuracy, but also because they scaled efficiently. Since tokens are processed in parallel, Transformers can fully utilise modern hardware such as GPUs. As models grew larger and were trained on increasingly vast datasets, new capabilities emerged, including fluent writing, conversation, and coding, without the need for explicit rule-based programming. Up to this point, we have focused on architecture—the structure of the system—but an equally important component is how the model is trained.

A Transformer does not inherently understand language; it learns through a process known as next-token prediction. During training, the model is exposed to billions of text sequences and asked to predict the next token. For instance, given the phrase “The board approved the annual _____”, it may predict “budget” or “report.” This prediction is compared to the actual word, and the difference is measured using a loss function. The resulting error is propagated back through the network using backpropagation, adjusting millions or even billions of parameters to reduce future errors. Over time, this process tunes embeddings, attention weights, and internal representations to capture the statistical structure of language.

This highlights a critical point: architecture defines capacity, while training defines capability. The model’s ability to preserve nuance, resolve references, and generate coherent outputs emerges from large-scale training rather than being explicitly programmed. The breakthrough after 2017 was not just the introduction of a new structure, but the combination of parallel Transformer architecture, self-attention, massive training data, and large-scale optimisation. Together, these made long-form reasoning practical at enterprise scale. While limitations remain—such as finite context windows and the absence of guaranteed factual accuracy—the improvement in reliability and scalability explains why generative AI has transitioned from research environments into real-world business applications.

Video 5: Training Objective and Inference Loop

To understand how large language models operate in both business and consumer environments, it is helpful to distinguish between two key phases: training and inference. Training is the phase in which the model learns patterns from data, while inference is the phase in which it generates responses during user interaction. Although the underlying mathematics is complex, the core logic behind both stages is relatively straightforward.

Large language models are trained using a method known as self-supervised learning. Rather than relying on labelled question-and-answer pairs, the model learns by predicting missing or next tokens in text. A token is a small unit of language, often a word or part of a word. For instance, when presented with the sentence “The finance team approved the annual _____,” the model learns that “budget” is a highly probable continuation. This prediction process is repeated billions of times across vast datasets, and each incorrect prediction leads to small adjustments in the model’s internal parameters, gradually improving its ability to model language.

This approach was applied at scale when Google rebuilt Google Translate using Transformer-based models after 2017. Earlier systems processed sentences sequentially, often resulting in awkward phrasing and a loss of context in longer inputs. With the shift to Transformer models trained on next-token prediction objectives, translation quality improved significantly. Internal evaluations showed improvements of over 50 per cent for several major language pairs, measured using the BLEU score, a standard metric that compares machine-generated translations with high-quality human references. These gains were achieved not through explicit grammar rules or dictionaries, but by learning statistical patterns from large volumes of translated text.

Once training is complete, the model moves into the inference phase. This is the stage where users interact with the system, such as when entering text into Google Translate. During inference, the model no longer updates its parameters. Instead, it generates outputs token by token using an autoregressive loop, where each predicted token is fed back into the sequence to generate the next. This explains why shorter inputs are processed quickly and consistently, while longer sentences tend to preserve meaning more effectively than earlier systems, especially when context plays a critical role.

However, the probabilistic nature of this process also introduces limitations. In cases involving ambiguity or highly technical language, the model may produce fluent but incorrect outputs. This is because it predicts likely sequences rather than verifying factual accuracy. Temperature, a parameter that controls randomness, also plays a role. Lower temperature settings produce more stable and literal outputs, which is desirable for tasks like translation, while higher settings introduce variation, which may be useful in creative applications but less appropriate where precision is required.

From a business perspective, this example highlights an important principle: the effectiveness of large language models stems from a simple training objective applied at scale, combined with rapid probabilistic inference. However, because outputs are based on likelihood rather than certainty, safeguards and human oversight remain essential, particularly in high-stakes contexts. In summary, these models learn by predicting the next token across vast datasets and generate responses through fast autoregressive loops, enabling powerful capabilities while still requiring careful and informed use.

Video 6: Why Bigger Models Suddenly “Reason”

One of the most counterintuitive discoveries in modern AI is that increasing model size does not simply lead to incremental improvement. At certain points, models begin to demonstrate capabilities that were previously absent or unreliable. This phenomenon is explained through scaling laws and emergent abilities. Scaling laws are empirical observations showing that model performance improves predictably as three factors increase together: the number of parameters, the volume of training data, and the amount of compute used during training. Parameters are the internal numerical weights the model learns, data refers to the text and code it is trained on, and compute is the processing power required to carry out training.

When researchers examined performance against model size, they observed smooth and consistent improvement curves. Doubling the number of parameters did not produce erratic outcomes; instead, it resulted in predictable reductions in error. This shifted the direction of AI development. Rather than focusing solely on new architectures, teams recognised that scaling proven architectures, particularly Transformers, could reliably enhance performance. However, what proved more surprising was that some capabilities did not improve gradually but appeared suddenly once models crossed certain thresholds. These are known as emergent abilities.

Smaller models can handle basic tasks such as sentence completion or short summaries, but they often struggle with multi-step reasoning, nuanced instruction following, and maintaining coherence in longer outputs. At larger scales, these same tasks become significantly more reliable. The shift can seem abrupt, with one version failing consistently while a slightly larger model succeeds. For example, in an enterprise setting where a services organisation tests models for proposal drafting, smaller models may generate individual sections effectively but fail to produce a complete, coherent document. They may contradict earlier assumptions, lose track of defined terms, or ignore formatting constraints.

In contrast, a larger model tested under the same conditions can maintain consistency across sections, reference earlier assumptions, and follow multi-step instructions with far fewer errors. The workflow remains unchanged; only the scale of the model differs. This ability to sustain coherence across long, structured outputs emerges once the model reaches a certain size. Similar patterns are observed across domains. Larger models are better at reasoning through multi-step problems, adhering to complex instructions, generating structured code, and even recognising when to use external tools such as calculators or document retrieval systems.

One way to understand this is through the idea of resolution. Smaller models capture coarse patterns in language, while larger models represent finer-grained relationships, including long-range dependencies and logical connections across extended text. However, scaling introduces diminishing returns. Each improvement requires significantly more compute, specialised hardware, and energy, making large-scale training expensive and resource-intensive. As a result, only a limited number of organisations can build frontier-scale models, and these models tend to dominate tasks where reasoning, synthesis, and long-context coherence are critical.

From a business perspective, this does not imply that larger models are always the best choice. Many applications can be handled effectively by smaller or fine-tuned models, particularly when tasks are well-defined. Frontier models provide the greatest value in scenarios where emergent abilities lead to meaningful performance gains. In summary, scaling laws demonstrate that increasing parameters, data, and compute improves performance in a predictable way, while emergent abilities explain why larger models often feel qualitatively different rather than just incrementally better. Understanding where these capabilities matter is essential for deploying generative AI both effectively and efficiently.

Video 7: Frontier Models

By 2025–2026, generative AI has moved decisively from experimentation into core enterprise infrastructure. For business leaders, the question is no longer whether to adopt AI, but which models to trust, where to deploy them, and how to govern them responsibly. Model selection has become a strategic decision, with direct implications for cost, risk, performance, and long-term dependency.

At the frontier of this ecosystem are a small number of highly capable model families. On the closed-model side, providers such as OpenAI, Anthropic, and Google lead in advanced reasoning, long-context understanding, and multimodal capability. These models benefit from significant training investment and continuous iteration, often delivering strong out-of-the-box performance. In parallel, open models from organisations such as Meta and Mistral AI have matured into enterprise-grade alternatives, offering greater control over deployment, data residency, and customisation—particularly valuable in regulated environments.

This creates a clear executive trade-off. Closed models typically offer superior capability and ease of use, but they introduce reliance on external providers for pricing, updates, and governance policies. Open models provide flexibility and control, but require internal technical expertise and may lag slightly in cutting-edge reasoning performance. To navigate this choice, organisations often turn to benchmarks, which measure capabilities such as reasoning, coding accuracy, and multimodal understanding. While benchmarks provide useful signals, they do not fully reflect real-world business performance.

In practice, what matters most is how models perform within actual workflows. For instance, in contract review, two models with similar benchmark scores may perform very differently when analysing long, complex agreements with cross-references and annexures. Models with stronger long-context reasoning are more likely to capture critical clauses and produce consistent summaries. Similarly, factors such as cost and scalability become significant at enterprise level, where token-based pricing can accumulate quickly in high-volume use cases like call summarisation or document processing. Latency also plays a role, with user-facing applications requiring faster responses than back-office systems.

Privacy, compliance, and governance are often decisive considerations. In industries such as finance, healthcare, and legal services, control over where data is processed and stored may outweigh marginal improvements in model capability. Organisations also require predictable behaviour, auditability, and alignment with internal risk standards. At the same time, multimodal capabilities are becoming increasingly important, as modern models can process text, images, and audio together. This enables more realistic business applications, such as analysing scanned documents, summarising customer interactions, and extracting insights from complex, unstructured data sources.

The key takeaway for leaders is that success with generative AI does not come from simply choosing the most powerful model available. Instead, it depends on selecting models that align with specific business workflows, operational constraints, and risk tolerance. Organisations that treat model selection as a strategic capability—rather than a purely technical decision—are best positioned to unlock sustained value from generative AI.

Video 8: Final Thoughts

As we close this week, it is helpful to connect the key ideas we have explored. We began by distinguishing between predictive AI and generative AI. Predictive systems focus on classifying or forecasting outcomes, while generative systems create new content. This distinction explains why

tools such as Smart Compose in Gmail can draft complete sentences as you type—it is not predicting a category, but generating language based on learned patterns, which makes the output feel natural and context-aware.

We then examined the core mechanism behind generation: autoregressive next-token prediction. Every response is produced one token at a time, with each token selected based on probability. What appears to be reasoning is, in reality, a sequence of statistically likely steps unfolding in real time. This understanding helps demystify how generative AI produces coherent and contextually relevant outputs.

To better understand cost and operational constraints, we explored tokenisation. Text is broken into tokens, which directly affect pricing, response time, and the amount of context a model can process. This is why longer inputs and richer context increase computational cost and latency, particularly in enterprise-scale deployments where efficiency becomes critical.

We also discussed the architectural breakthrough that enabled modern generative AI: the Transformer and its self-attention mechanism. Self-attention allows models to understand relationships across entire sentences and documents, rather than processing text sequentially. This shift explains why systems such as Google Translate improved significantly after moving from earlier RNN-based approaches to Transformer-based architectures, resulting in more fluent and contextually accurate outputs.

Next, we looked at training and inference. Models are trained using self-supervised next-token prediction across massive datasets, gradually learning the structure and patterns of language. During inference, they generate responses using probabilistic methods, with parameters that balance creativity and consistency depending on the use case.

The key takeaway is clear: generative AI is not magic. It is the result of architecture, data, scale, and probability working together. Understanding these foundations allows organisations to move beyond hype and use AI as a meaningful source of competitive advantage.